Waves in Two Dimensions

So far, we have only talked about waves in 1 dimension. A wave moves along a string, or a planar sound wave moves in a single direction. Things got a bit more complicated when we added in a change in medium/wave speed. The behavior of waves gets *vastly* more complicated and/or rich and interesting when you look at them in two dimensions. Even though the behavior is a lot more complicated, we'll still get the same behavior for sound, light, and water waves. It remains true that if you've seen one wave, you've seen them all.

These wave effects can matter tremendously in music. Why is vibrato so widely used? What happens if you wire a speaker backwards? Looking at the properties of 2-D waves can help us understand these questions.

1. Single Source

Picture a pebble tossed into a pond. You get circular ripples moving outwards. The amplitude of the wave does decrease with distance, but we'll ignore that for now. If we have a single source, then the wave behavior will look like

$$f(r - vt) \tag{1}$$

Equation 1 looks very much like the wave equations we're used to $f(x - vt)$, but we've replaced x with r where r is the distance from the soure to a given point in 2-dimensional (or higher!) space. Because every point in space with the same value of r will behave the same way, and points with constant distance from a source form circles, our wave will look like circular patterns moving away from the source. If we do want to worry about waves weakening with distance, then Equation 1 turns into

$$\frac{r^\alpha}{f}(r - vt) \tag{2}$$

where α depends on how many dimensions the wave is moving in. For 3D, $\alpha = -1$, that is, wave amplitude/pressure fall off as $\frac{1}{r}$.

Unfortunately Desmos is not great at the sort of 2D plots we'll want to look at here, but you're welcome to look at the python script waves_2d.py in the python directory on github. That will make 2D animations showing waves spreading outwards from a variety of point setups. If you uncomment the line
pos=[[0,0]]
the script will show a sine wave moving away from a source at $[0, 0]$. I haven't worried about the amplitude decay; as we'll see, our hands will be full enough without that (and the amplitude usually doesn't matter much anyways).

2. Two Sources, In-Phase

If we have two sources, the behavior gets a lot more complicated. Again, picture a pond but this time you threw in two pebbles. Each pebble makes a set of rings, but now some places the rings

reinforce each other, and some places they cancel out. If we ask "what happens at point (x, y) when we have a source of waves at (x_1, y_1)?", we first need to note that

$$r = \sqrt{(x - x_1)^2 + (y - y_1)^2} \tag{3}$$

Once we have calculated r, which does *not* change with time, then we can go look up our wave height for any time t by looking at Equation 1. If we have a sine wave, then we have our usual situation

$$\sin(kr - \omega t) \tag{4}$$

where as usual the wave speed $v = \frac{omega}{k}$. Plugging in Equation 3 for r into Equation 4 will let us work out the height of our wave given a point (x, y).

What happens when we add a second source? The wave equation (at least in the cases we'll consider) stays linear. That is, we just add the two waves together. If our sources are at (x_1, y_1) and (x_2, y_2) and r_1 and r_2 are the distances from our chosen point (x, y) and the two sources, our sine wave now becomes

$$\sin(kr_1 - \omega t) + \sin(kr_2 - \omega t) \tag{5}$$

To try and make more sense of this, we'll introduce a transform. Let $r = \frac{r_1 + r_2}{2}$ and $dr = \frac{r_1 - r_2}{2}$. In words, r is the distance from our point to the *center* of the two sources, and dr is the difference in distance from one point relative to the center. If we add the two equations together we get

$$r + dr = r_1/2 + r_1/2 = r1 \tag{6}$$

and

$$r - dr = \frac{r_1 + r_2}{2} - \frac{r_1 - r_2}{2} = r_2 \tag{7}$$

We can plug Equations 6 and 7 into Equation 5 to get:

$$\sin(k(r + dr) - \omega t) + \sin(k(r - dr) - \omega t) \tag{8}$$

We're now going to use $\sin(a + b) = \sin(a)\cos(b) + \cos(a)\sin(b)$ with $b = dr$ to get

$$\sin(kr - \omega t + kdr) + \sin(kr - \omega t - kdr) \tag{9}$$

$$= \sin(kr - \omega t)\cos(kdr) + \cos(kr - \omega t)\sin(kdr) + \sin(kr - \omega t)\cos(-kdr) + \cos(kr - \omega t)\sin(-kdr) \tag{10}$$

$$= 2\sin(kr - \omega t)\cos(kdr) \tag{11}$$

Equation 11 should look rather familiar. Mathematically, it's the same as the beating from two closely-spaced frequencies, but the interpretation is now rather different. The first term $\sin(kr - \omega t)$ is just a sine wave coming from the midpoint of the two sources.

The second term in Equation 11 is an overall amplitude. If we pick dr to be very small - that is, we're at a point in space where the *difference* in distance between our point an the two sources is very small, then kdr is very small, and $\cos(kdr)$ is very close to one. If we're at a point that is the same distance from our two sources, then their waves *add* and we get a *strong* signal. This is called *constructive interference* - the two waves are adding *in-phase*. What happens if $kdr = \pi/2$? Now $\cos(kdr) = 0$, and the waves are cancelling each other out. We don't get a signal, and this is

called *destructive interference*. If we convert from k to wavelength λ then we see we get destructive interference at $dr = \lambda/4$. Recall that we defined dr to be the change in distance between the midpoint and each source, so the difference in distance is $\lambda/2$. That is, if we are exactly half a wavelength further from one source than another, they destructively interfere.

We get constructive interference not just for $dr = 0$. Whenever the distance difference is an integer number of wavelengths we'll get constructive interference.

3. Pattern from Two Sources

Let's consider two wave sources. We'll put the first at the origin $(0, 0)$ and the second at a distance d along the $y-$axis, so its position is $(0, d)$. We want to find the distances from the two sources to a receiver that's a distance r from the origin that makes an angle θ with the $x-$axis. The position of the receiver is then $(r\cos(\theta), r\sin(\theta))$. The distance from the first source to the receiver is obviously r (after all, that's why we put the first source at the origin). Now we want to work out the distance from the receiver to the second source. Recall that the distance between two points separated by $(\delta x, \delta y)$ is $\sqrt{\delta x^2 + \delta y^2}$. The $x-$separation is just $r\cos(\theta)$ since source 2 is at $x = 0$, but the $y-$separation is $r\sin(\theta) - d$. We then get the distance between source 2 and the receiver r_2 is:

$$r_2 = \sqrt{(r\cos(\theta))^2 + (r\sin(\theta) - d)^2} \tag{12}$$

We can expand the second term in the equation since $(a - b)^2 = a^2 - 2ab + b^2$ to get:

$$r_2 = \sqrt{r^2\cos(\theta)^2 + r^2\sin(\theta)^2 - 2r\sin(\theta)d + d^2} \tag{13}$$

Since $\cos^2 + \sin^2 = 1$ we can combine the first terms to get

$$r_2 = \sqrt{r^2 - 2r\sin(\theta)d + d^2} \tag{14}$$

We can pull a factor of r^2 out of the square root by multiplying/dividing the second term by r and the third term by r^2 to get:

$$r_2 = \sqrt{r^2\left(1 - 2\frac{\sin(\theta)d}{r} + \frac{d^2}{r^2}\right)} \tag{15}$$

$$r_2 = r\sqrt{1 - 2\frac{\sin(\theta)d}{r} + \frac{d^2}{r^2}} \tag{16}$$

This expression is exactly correct, but can be awkward to work with. We can make some progress if we assume we're a long ways from the sources, so that r is much, much larger than d. If that's the case, then the second term, which has a factor of $\frac{d}{r}$ will be a lot smaller than 1, and the third term, which is $\frac{d^2}{r^2}$ will not only be a lot smaller than 1, it will also be a lot smaller than the second term.

At this point, we'll want to know what $\sqrt{1 + \delta}$ is approximately equal to when δ is very small. We can figure this out by looking at $(1 + \epsilon)^2$ where ϵ is also a very small number. We know that

$(1+\epsilon)^2 = 1+2\epsilon+\epsilon^2$, but if ϵ is small, then ϵ^2 is tiny. For instance, if $\epsilon = 0.001$, then $\epsilon^2 = 0.000001$. If we throw away the extremely tiny ϵ^2 term, then we have

$$(1+\epsilon)^2 \simeq 1+2\epsilon \tag{17}$$

Now we take a square root of both sides and we have

$$1+\epsilon \simeq \sqrt{1+2\epsilon} \tag{18}$$

Finally, we define $\delta = 2\epsilon$, and after swapping the left and right sides of the equation, we get

$$\sqrt{1+\delta} \simeq 1+\frac{\delta}{2} \tag{19}$$

This gives us the very useful result that the square root of 1 plus a small number is 1 plus half of that small number. We can apply this to Equation 16 to get the approximate distance when r is much larger than d:

$$r_2 \simeq r\left(1-\frac{d\sin(\theta)}{r}\right) = r - d\sin(\theta) \tag{20}$$

The *difference* in distances is

$$r - r_2 = r - (r - d\sin(\theta)) = d\sin(\theta) \tag{21}$$

Equation **??** tells us something remarkably simple: if we are a long ways from the sources, the difference in distance between the two sources and our receiver is just $d\sin(\theta)$ where again d is the distance between the two sources, and θ is the angle our receiver makes with the line from the first source that is perpendiculat to the line connecting the two sources.